These are notes for Malek Abdesselam's Functional Analysis class for Fall 2026.

Contents

Motivation

Consider a space V of (sufficiently nice) functions $f: \mathbb{R} \to \mathbb{R}$. Then, similar to how a vector in $\mathbb{R}^n$ can be denoted as

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},$$

we may view the function $f: \mathbb{R} \to \mathbb{R}$ as a "column vector" with uncountable index. That is, $f = (f(x))_{x\in\mathbb{R}}$. If $L\colon V \to V$ is a linear map, then an analogue of the finite-dimensional matrix representation can be viewed as a kernel that f is integrated against; that is,

$$(Lf)(x) = \int_{\mathbb{R}} K(x,y)f(y)\,dy.$$

In the special case that $L = \mathrm{id}_V$, then the kernel $K(x,y)$ is given by the Dirac delta distribution, $K(x,y) = \delta(x-y)$. The space of these distributions is the primary object of study in Fourier analysis.

Locally Convex Topological Vector Spaces

Recall that if X is a set, then $\mathcal{T} \subseteq \mathcal{P}(\mathcal{P}(X))$ is a topology on X if it satisfies:

- $\emptyset, X \in \mathcal{T}$;
- $U_1, U_2 \in \mathcal{T}$ implies $U_1 \cap U_2 \in \mathcal{T}$;
- $\{U_i : i \in I\} \subseteq \mathcal{T}$ implies $\bigcup_{i\in I} U_i \in \mathcal{T}$.

Definition: **Definitions**

We say $(V, \mathcal{T})$ is a *topological vector space* (over $\mathbb{K}$) if V is a $\mathbb{K}$-vector space whose topology is such that the maps

$$\begin{aligned} V \times V &\to V \\ (u,v) &\mapsto u+v \end{aligned}$$

and

$$\begin{aligned} \mathbb{K} \times V &\to V \\ (\lambda,v) &\mapsto \lambda v \end{aligned}$$

are continuous with respect to the product topology. Note that $\mathbb{K}$ is equipped with its traditional topology.

The morphisms of topological vector spaces are the *continuous* linear maps. We say $V \cong W$ are isomorphic as topological vector spaces if there is $\varphi\colon V \to W$ that is bijective, linear, and continuous, and φ^{-1} is also continuous.

Definition: Let V be a $\mathbb{K}$-vector space. We say $\rho\colon V \to [0,\infty)$ is a *seminorm* if

$$\begin{aligned} \rho(v+w) &\leq \rho(v) + \rho(w) \\ \rho(\lambda v) &= |\lambda|\rho(v) \end{aligned}$$

for any $v, w \in V$ and $\lambda \in \mathbb{K}$.

Remark: The only difference between a seminorm and a norm is that $\rho(v) = 0$ need not imply that $v = 0$. A norm is a seminorm.

We will let $\mathrm{SN}(V)$ be the set of all seminorms of V.

Definition: Let V be a $\mathbb{K}$-vector space. Let $\mathcal{N} \subseteq \mathrm{SN}(V)$. Define $\mathcal{T}(\mathcal{N})$ to be a topology with basic open sets $U(x,F,\varepsilon) \subseteq V$ given by

$$\begin{aligned} U(x,F,\varepsilon) &= \{y \in V : \rho(y-x) < \varepsilon \text{ for all } \rho \in F\} \\ &= \bigcap_{\rho \in F} U(x,\rho,\varepsilon). \end{aligned}$$

for any $x \in V$, $F \subseteq \mathcal{N}$ finite, and $\varepsilon > 0$.

Remark: I prefer to use the subbasic open sets

$$U(x,\rho,\varepsilon) = \{y \in V : \rho(y-x) < \varepsilon\}$$

and write intersections.

We observe that the subsets of the form $U(x,\{\rho\},\varepsilon)$ as ρ ranges across $\mathcal{N}$ forms a subbase for the topology of V.

Proposition: If $F \subseteq \mathcal{N}$ is finite, $\varepsilon > 0$, and $x \in V$, then

$$U(x,F,\varepsilon) \in \mathcal{T}(\mathcal{N}).$$

Proof. Suppose $y \in U(x,F,\varepsilon)$. Our goal is to find $\eta > 0$ such that $U(y,F,\eta) \subseteq U(x,F,\varepsilon)$. Suppose $z \in U(y,F,\eta)$. If $\rho \in F$, then

$$\begin{aligned} \rho(z-x) &\leq \rho(z-y) + \rho(y-x) \\ &< \eta + \rho(y-x). \end{aligned}$$

Therefore, we will define η to be small enough such that $\eta + \max_{\rho \in F} \rho(y-x) < \varepsilon$. This gives the desired result. □

Definition: A topological space is called a locally convex topological vector space (LCTVS) if the topology is generated by a family of seminorms $\mathcal{N}$ with the property that

$$\bigcap_{\rho \in \mathcal{N}} \{v : \rho(v) = 0\} = \{0\}.$$

A family $\mathcal{N}$ of seminorms that satisfies this property is said to be *separating*.

Example: If H is a Hilbert space with orthonormal basis $\{e_n : n \in \mathbb{N}\}$, then the seminorms $v \mapsto |\langle v, e_i \rangle|$ are separating.

Theorem: If V is a $\mathbb{K}$-LCTVS and $\mathcal{N}$ is a collection of seminorms that defines the topology on V, then the pair $(V, \mathcal{T}(\mathcal{N}))$ is Hausdorff if and only if $\mathcal{N}$ separates the points of V.

Proof. Assume $\mathcal{N}$ separates the points of V. Suppose $v \neq w$ are distinct elements of V. Then, since $v - w \neq 0$, there is some $\rho \in \mathcal{N}$ such that $\rho(v-w) = \varepsilon_0 > 0$. Consider the finite set $F = \{\rho\}$, and choose some $0 <$

$\varepsilon < \varepsilon_0/2$. Then, we claim that the basic open sets $U(v,F,\varepsilon)$ and $U(w,F,\varepsilon)$ are disjoint. This follows from the fact that if $x \in U(v,F,\varepsilon)$, then $\varepsilon_0 = \rho(v-w) \leq \rho(v-x) + \rho(x-w)$, so that $\rho(w-x) \geq \varepsilon_0 - \varepsilon > \varepsilon_0 - \varepsilon_0/2 > \varepsilon$, yielding the desired result.

Now, suppose $\mathcal{N}$ does not separate the points of V. Then, there are some distinct $v \neq w$ such that for all $\rho \in \mathcal{N}$, we have $\rho(v-w) = 0$. In particular, observe that for all finite subsets $F \subseteq \mathcal{N}$ and all $\varepsilon > 0$, we have that w is contained in $U(v,F,\varepsilon)$, meaning that there are no basic open sets that separate v and w. Since any open subset of V contains a basic open subset, it follows that for any open subsets $v \in U$ and $w \in U'$, we have some finite F such that $U(v,F,\varepsilon) \subseteq U$, but this also has $w \in U(v,F,\varepsilon)$, implying that $U \cap U' \neq \emptyset$. Thus, V is not Hausdorff. $\square$

Examples

Remark: If we remove the assumption that the family $\mathcal{N}$ is separating, then the topology induced by the trivial seminorm $\rho\colon V \to [0,\infty)$ is the indiscrete topology on V. Similarly, there is a family of seminorms that induces the discrete topology on V.

(i) Let $(V, \|\cdot\|)$ be a normed vector space over $\mathbb{K}$. Examples of these include Banach spaces such as $L^p(X,\Sigma,\mu)$, where Σ is a σ-algebra and μ is a positive measure on the measurable space (X,Σ), and $p \in [1,\infty]$.

There is a metric on this normed space $d(x,y) = \left\|x-y\right\|$, which gives rise to the metric topology $\mathcal{T}(d)$. We have $U \in \mathcal{T}(d)$ if and only if $U \subseteq V$ is such that for any $x \in U$, there exists $\varepsilon > 0$ such that

$$\begin{aligned} \{y \in V : d(x,y) < \varepsilon\} &=: U(x,\varepsilon) \\ &\subseteq U. \end{aligned}$$

This is a locally convex space with seminorm $\mathcal{N} = \{\|\cdot\|\}$. There are two possibilities for finite subsets $F \subseteq \{\|\cdot\|\}$, meaning that if $F = \emptyset$, then $U(x,F,\varepsilon) = U(x,\emptyset,\varepsilon) = V$, while if $F = \{\|\cdot\|\}$, then

$$\begin{aligned} U(x,F,\varepsilon) &= \{y \in V : \rho(x-y) < \varepsilon \text{ for all } \rho \in F\} \\ &= \left\{y \in V : \left\|x-y\right\| < \varepsilon\right\} \\ &= U(x,\varepsilon). \end{aligned}$$

(ii) Define $s(\mathbb{N},\mathbb{K})$ to be the space of sequences $x = (x_n)_{n\geq 0}$ in $\mathbb{K}^{\mathbb{N}}$ such that for all $k \geq 0$, $\sup_{n\geq 0}\langle n\rangle^k |x_n| < \infty$, where $\langle n\rangle = \sqrt{1+n^2}$ is the bracket. This is the space of *rapidly decaying sequences* in $\mathbb{K}$. In essence, elements of $s(\mathbb{N},\mathbb{K})$ decay faster than any inverse power of the index n.

We define the seminorms

$$\rho_k(x) = \sup_{n\geq 0}\langle n\rangle^k |x_n|.$$

In fact, the $\{\rho_k : k \in \mathbb{N}\}$ are norms. Let

$$\mathcal{N} := \{\rho_k : k \in \mathbb{N}\}$$

We obtain a locally convex on $s(\mathbb{N},\mathbb{K})$ with topology induced by the family $\mathcal{N}$.

Later, we will see that $(s(\mathbb{N},\mathbb{K}), \mathcal{T}(\mathcal{N}))$ is *not* a normed space. That is, there does not exist a norm $\|\cdot\|$ such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\{\|\cdot\|\})$.

(iii) Take $V = C([0,\infty),\mathbb{R})$ to be the set of all real-valued continuous functions on $[0,\infty)$ with pointwise addition and scalar multiplication. This is a real vector space.

At first, we may think $\|f\| = \sup_{t\in\mathbb{R}}|f(t)|$ would define a norm, but this is not satisfactory since the functions in V are not necessarily bounded.

Therefore, we want to restrict to compact subsets. Take

$$\rho_k(f) := \sup_{0\leq t\leq k} |f(t)|.$$

Notice that the ρ_k are not norms (use Urysohn's Lemma or your favorite piecewise function to construct one). However, the family $\mathcal{N} = \{\rho_k : k \geq 0\}$ is a separating family.

Consider the set of functions $V_0 = \{f \in V : f(0) = 0\}$. Then, $V_0 \leq V$ is a closed subset.

Proof. Consider the linear functional $\mathrm{ev}_0 \colon V \to \mathbb{R}$ given by $f \mapsto f(0)$. We claim that this linear functional is continuous.

Let $U \subseteq \mathbb{R}$ be open. Let $z \in U$, and let $\varepsilon > 0$ be such that $U(z,\varepsilon) := \left\{y \in \mathbb{R} : \left|z-y\right| < \varepsilon\right\}$ is contained in U. Let $f \in \mathrm{ev}_0^{-1}(U(z,\varepsilon))$, and set $\varepsilon_0 = |z - f(0)|$. To show that $\mathrm{ev}_0^{-1}(U(z,\varepsilon))$ is open, it suffices to find some seminorm ρ_k and some $\delta > 0$ such that $U(f,\rho_k,\delta) \subseteq \mathrm{ev}_0^{-1}(U(z,\varepsilon))$.

In fact, we may choose $k = 1$ and $\delta = \varepsilon - \varepsilon_0$. Suppose $g \in U(f,\rho_1,\delta)$. Then, we have that

$$\begin{aligned}
|g(0) - z| &\leq |g(0) - f(0)| + |f(0) - z| \\
&\leq \sup_{t\in[0,1]} |g(t) - f(t)| + \varepsilon_0 \\
&< \delta + \varepsilon_0 \\
&= \varepsilon.
\end{aligned}$$

In particular, Thus, $U(f,\rho_1,\delta) \subseteq \mathrm{ev}_0^{-1}(U(z,\varepsilon))$, so that $\mathrm{ev}_0^{-1}(U(z,\varepsilon))$ is open as any element in this set admits a basic open neighborhood contained in the set. Therefore, ev_0 is continuous.

Since $V_0 = \mathrm{ev}_0^{-1}(\{0\})$, it follows that V_0 is closed since $\{0\} \subseteq \mathbb{R}$ is closed. □

We claim that the subspace topology on V_0 is given by $\mathcal{T}_0 = \mathcal{T}\left(\left\{\rho_k|_{V_0} : k \geq 0\right\}\right)$.

Proof. First, we will show that, given a subbasic open neighborhood $U = V_0 \cap U(x,\rho_k,\varepsilon)$ in the subspace topology of V_0 and any element y in this subbasic open neighborhood, there is an open neighborhood $V \in \mathcal{T}\left(\left\{\rho_k|_{V_0} : k \geq 0\right\}\right)$ such that such that $y \in V \subseteq U$.

If $y \in U$, we observe that $y \in U(x,\rho_k,\varepsilon)$. In particular, since these form a subbase for $\mathcal{T}(\{\rho_k : k \geq 0\})$, it follows that there is some finite collection $F = \left\{\rho_{i_j} : j = 1,\dots,m\right\}$ of seminorms and some $\delta > 0$ such that

$$U(y,F,\delta) = \bigcap_{j=1}^{m} U\left(y,\rho_{i_j},\delta\right).$$

Since $U\left(y,\rho_{i_j}|_{V_0},\delta\right) \subseteq U\left(y,\rho_{i_j},\delta\right)$ for each j, it follows that we have

$$\bigcap_{j=1}^{m} U\left(y,\rho_{i_j}|_{V_0},\delta\right) \subseteq V_0 \cap \left(\bigcap_{j=1}^{m} U\left(y,\rho_{i_j},\delta\right)\right),$$

whence setting V to equal the left hand side yields the desired result. Therefore, $\mathcal{T}\left(\left\{\rho_k|_{V_0} : k \geq 0\right\}\right)$ refines the subspace topology $\mathcal{T}_0$.

In the reverse direction, observe that if $y \in U\left(x,\rho_j|_{V_0},\varepsilon\right)$ in V_0, then $y \in U\left(x,\rho_j,\varepsilon\right)$, so there is some $\delta > 0$ and finite $F = \left\{\rho_{i_j} : j = 1,\dots,m\right\}$ such that

$$U(y,F,\varepsilon) = \bigcap_{j=1}^{m} U\left(y,\rho_{i_j},\delta\right)$$

$$\subseteq U\left(x, \rho_j, \varepsilon\right),$$

but we notice then that $U\left(x, \rho_j, \varepsilon\right) \cap V_0 \subseteq U\left(x, \rho_j|_{V_0}, \varepsilon\right)$, so that $V_0 \cap U(y, F, \varepsilon) \subseteq U\left(x, \rho_j|_{V_0}, \varepsilon\right)$, so that $\mathcal{T}_0$ refines $\mathcal{T}\left(\left\{\rho_k|_{V_0} : k \geq 0\right\}\right)$. Thus, these topologies are equal. □

We may consider $\mathcal{B}(\mathcal{T}_0)$ to be the Borel σ-algebra on V_0. Brownian motions are a special class of probability measure on $(V_0, \mathcal{B}(\mathcal{T}_0))$. Specifically, they are the probability measures such that for every $n \geq 0$, partition $0 < t_1 < t_2 < \cdots < t_n$, the random variables

$$\begin{aligned}
X_1 &= f \mapsto f(t_1) \\
X_2 &= f \mapsto (f(t_2) - f(t_1)) \\
X_3 &= f \mapsto (f(t_3) - f(t_2)) \\
&\vdots \\
X_n &= f \mapsto (f(t_n) - f(t_{n-1}))
\end{aligned}$$

are independent centered Gaussian random variables with variances $t_1, t_2 - t_2, \dots, t_n - t_{n-1}$ for an arbitrary f.

Similar to the case of $s(\mathbb{N}, \mathbb{K})$, the space $(V_0, \mathcal{T}_0)$ is not a normed space.

(iv) Consider the vector space $V = \mathbb{R}^{\mathbb{N}}$, which is the space of all real-valued sequences. Let $\mathcal{T}$ be the product topology on $\mathbb{R}^{\mathbb{N}}$. We are curious as to whether $(V, \mathcal{T})$ is a locally convex topological vector space.

Consider the function $d\colon V \times V \to [0, \infty)$ defined by

$$d(x, y) = \sum_{n=0}^{\infty} 2^{-n} \min\left(1, \left|x_n - y_n\right|\right).$$

Then, we have that d is a distance metric, and $\mathcal{T} = \mathcal{T}(d)$.

Proof. We start by showing that if $x \in V$ is any element and $\varepsilon > 0$, then $U(x, \varepsilon)$ is open in $\mathcal{T}$. Translating, we take $x = 0$ (since the metric d is translation-invariant), so that we are now seeking to show that $U(0, \varepsilon)$ is open. This is equivalent to showing that if $x \in U(0, \varepsilon)$, then there is a finite collection of subbasic open sets $\pi_j^{-1}\left(U_j\right)$ whose intersection W contains x and is contained in $U(0, \varepsilon)$.

To start, we observe that there is some index n_0 such that

$$\sum_{n=n_0+1}^{\infty} 2^{-n} < \varepsilon/2.$$

We then select $U_j = \left(x_j - \varepsilon/4, x_j + \varepsilon/4\right)$ for each $j = 0, \dots, n_0$. It then follows that $x \in W$, and we have for any $y \in W$ that

$$\begin{aligned}
d(x, y) &= \sum_{j=0}^{\infty} 2^{-j} \min\left(1, \left|x_j - y_j\right|\right) \\
&\leq \sum_{j=0}^{n_0} 2^{-j} \min\left(1, \left|x_j - y_j\right|\right) + \sum_{j=n_0+1}^{\infty} 2^{-j} \\
&< \sum_{j=0}^{n_0} 2^{-j} (\varepsilon/4) + \varepsilon/2 \\
&\leq \varepsilon,
\end{aligned}$$

whence $x \in W \subseteq U(0, \varepsilon)$.

Now, consider a basic open set $W = \bigcap_{j=0}^{n_0} \pi_j^{-1}\left(U_j\right)$. Let $x \in W$. Let $\delta_0, \dots, \delta_{n_0}$ be such that $\left(x_j - \delta_j, x_j + \delta_j\right) \subseteq U_j$ for each $j = 1, \dots, n$, and set $\delta = 2^{-n_1} \min\left(\delta_0, \dots, \delta_{n_0}\right)$, where n_1 is any integer strictly larger than n_0.

Notice then that if $y \in U(x, \delta)$, then we have for each $j = 0, \dots, n_0$ that

$$\begin{aligned}\left|x_j - y_j\right| &< 2^j \delta \\ &\leq 2^{n_0} 2^{-n_1} \min\left(\delta_0, \dots, \delta_{n_0}\right) \\ &< \min\left(\delta_0, \dots, \delta_{n_0}\right) \\ &\leq \delta_j.\end{aligned}$$

In particular, this means we must have $U(x, \delta) \subseteq W$.

Therefore, we have that $\mathcal{T}$ and $\mathcal{T}(d)$ generate the same topology on V. □

In particular, we may consider the seminorms

$$\rho_n(x) = |x_n|,$$

and $\mathcal{N} = \{\rho_n : n \geq 0\}$. Then, we claim that $\mathcal{T} = \mathcal{T}(\mathcal{N})$.

Proof. A basic open neighborhood of $x \in V$ in $\mathcal{T}(\mathcal{N})$ is given by the sets

$$U(x, F, \varepsilon) = \bigcap_{j \in F} \left\{(y_n)_n : \left|x_j - y_j\right| < \varepsilon\right\}.$$

Meanwhile, a basic open neighborhood of x in $\mathcal{T}$ is given by the sets

$$U'(x, F, \varepsilon) = \bigcap_{j \in F} \pi_j^{-1}\left(\left(x_j - \varepsilon, x_j + \varepsilon\right)\right).$$

In fact, these are the same subsets of V by the definition of the projection map $\pi_j \colon V \to \mathbb{R}$ taking $(x_n)_n \mapsto x_j$.

Therefore, if we have $y \in U(x, F, \varepsilon)$ in $\mathcal{T}(\mathcal{N})$, then if we define

$$\delta = \frac{1}{2} \min\left(\left\{\left|\varepsilon - y_j\right| : y_j \in F\right\}\right),$$

we have that $U'(y, F, \delta) \subseteq U(x, F, \varepsilon)$. Similarly, if $y \in U'(x, F, \varepsilon)$, then setting δ to equal the same as above, we have $U(y, F, \varepsilon) \subseteq U'(x, F, \varepsilon)$.

This gives $\mathcal{T} = \mathcal{T}(\mathcal{N})$. □

In particular, V is not a normed space.

(v) Consider $c_{00}(\mathbb{N}, \mathbb{K}) \cong \mathbb{K}[x]$ to consist of all the finitely supported sequences $(x_n)_{n \geq 0}$. This is a vector space. Consider the seminorms $\mathcal{N} = \mathrm{SN}(V)$, and the locally convex space $(V, \mathcal{T}(\mathcal{N}))$. Then, this is a locally convex space that is not metrizable. The topology $\mathcal{T}(\mathcal{N})$ is known as the finest locally convex topology.

General Properties of Topological Vector Spaces

Definition: Let $(V, \mathcal{T})$ be a $\mathbb{K}$-TVS, and let $\tau \in \mathrm{SN}(V)$. Then, τ is a *continuous seminorm* for $(V, \mathcal{T})$ if $\tau \colon V \to [0, \infty) \subseteq \mathbb{R}$ is continuous.

Proposition: Let V be a $\mathbb{K}$-vector space, $\mathcal{N} \subseteq \mathrm{SN}(V)$ define a locally convex topology on V. If $\tau \in \mathrm{SN}(V)$, then τ is continuous for $(V, \mathcal{T}(\mathcal{N}))$ if and only if there exists $C \geq 0$, $n \in \mathbb{N}$, and $\rho_1, \dots, \rho_n \in \mathcal{N}$ such that

$$\tau(v) \leq C \max\left\{\rho_j(v) : j = 1, \dots, n\right\} \tag{$*$}$$

or

$$\tau(v) \leq C \sum_{j=1}^{n} \rho_j(v).$$

Proof. Suppose τ is continuous. Then, $[0,1)$ is open in $[0,\infty)$, so $\tau^{-1}([0,1))$ is open in $\mathcal{T}(\mathcal{N})$ with $0 \in \tau^{-1}([0,1))$.

So, by definition, there exists $\varepsilon > 0$ and $\rho_1, \dots, \rho_n$ such that

$$\bigcap_{j=1}^{n} U\left(0, \rho_j, \varepsilon\right) \subseteq \tau^{-1}([0,\infty)).$$

That is, for any $i \in [n]$, we have that if $\rho_i(x) < \varepsilon$, then $\tau(x) < 1$.

Let $x \in V$ and $\lambda > 0$. Then,

$$\begin{aligned} \tau(x) &= \tau\left(\lambda\lambda^{-1}x\right) \\ &= \lambda\tau\left(\lambda^{-1}x\right) \\ &< \lambda \end{aligned}$$

if for every i, we have $\rho_i\left(\lambda^{-1}x\right) < \varepsilon$.

Define $J = \left\{\lambda > 0 : \rho_i\left(\lambda^{-1}x\right) < \varepsilon \text{ for all } i \in [n]\right\}$. Rewriting, we have

$$\begin{aligned} J &= \left\{\lambda > 0 : \frac{1}{\varepsilon}\rho_i(x) < \lambda \text{ for all } i \in [n]\right\} \\ &= \bigcap_{i=1}^{n} \left(\frac{1}{\varepsilon}\rho_i(x), \infty\right), \end{aligned}$$

giving $\inf(J) = \frac{1}{\varepsilon}\max\{\rho_i(x) : 1 \leq i \leq n\}$. Therefore, if $\lambda \in J$, then $\tau(x) \leq \lambda$, whence $\tau(x) \leq \inf(J)$. Therefore, we have shown that $\tau(x) \leq \frac{1}{\varepsilon}\max\{\rho_i(x) : 1 \leq i \leq n\}$.

In the reverse direction, suppose $(*)$ holds. Let $x \in V$, $\rho_1, \dots, \rho_n \in \mathcal{N}$, and let $\eta > 0$. Let

$$y \in \bigcap_{i=1}^{n} U(x, \rho_i, \eta).$$

By the reverse triangle inequality, we have

$$\begin{aligned} \left|\tau(y) - \tau(x)\right| &\leq \tau(y - x) \\ &< \varepsilon, \end{aligned}$$

but

$$\begin{aligned} \tau(y - x) &\leq C\max\{\rho_i(y - x) : 1 \leq i \leq n\} \\ &\leq C\eta \\ &< \varepsilon \end{aligned}$$

if η is small enough. Thus, we have for all $y \in \bigcap_{j=1}^{n} U\left(x, \rho_j, \eta\right)$, then $\left|\tau(y) - \tau(x)\right| < \varepsilon$, whence τ is continuous at x. □

Remark: If $\tau \in \mathcal{N}$, then τ is continuous with respect to $\mathcal{T}(\mathcal{N})$.

Proposition: Let $(V, \mathcal{T}(\mathcal{N}))$ be a LCTVS. Then, $U \subseteq V$ is open if and only if for all $x \in U$, there is some continuous seminorm τ and some $\varepsilon > 0$ such that $U(x, \tau, \varepsilon) \subseteq U$.

Theorem (Continuity of Multilinear Maps on LCTVSes): Suppose $n \geq 1$, and $V_1, \dots, V_n, W$ are $\mathbb{K}$-LCTVSes, and let $M\colon V_1 \times \cdots \times V_n \to W$ be a $\mathbb{K}$-multilinear map.

Then, M is (jointly) continuous if and only if for all continuous seminorm τ on W, there exist $\rho_1, \dots, \rho_n$ continuous seminorms on $V_1, \dots, V_n$ respectively such that for all tuples $(x_1, \dots, x_n) \in V_1 \times \cdots \times V_n$,

$$\tau(M(x_1, \dots, x_n)) \leq \rho(x_1) \cdots \rho(x_n). \tag{†}$$

Proof. Suppose M is (jointly) continuous. That is, M is equipped with the product topology. Since τ is a continuous seminorm, we have that τ is a continuous map from $W \to [0,\infty)$, we have $\tau^{-1}([0,1))$ is open in W. Moreover, $M(0,\dots,0) = 0 \in \tau^{-1}([0,1))$. Since M is continuous at $(0,\dots,0)$, there is a neighborhood $U_1 \times \cdots \times U_n$ in $V_1 \times \cdots \times V_n$ where each U_i is a neighborhood of 0 in V_i, such that $M(U_1 \times \cdots \times U_n) \subseteq \tau^{-1}([0,1))$.

Let $(x_1, \dots, x_n) \in V_1 \times \cdots \times V_n$, and let $\lambda_1, \dots, \lambda_n > 0$. We have

$$\begin{aligned}
\tau(M(x_1, \dots, x_n)) &= \tau\left(M\left(\lambda_1 \lambda_1^{-1} x_1, \dots, \lambda_n \lambda_n^{-1} x_n\right)\right) \\
&= \lambda_1 \cdots \lambda_n \tau\left(M\left(\lambda_1^{-1} x_1, \dots, \lambda_n^{-1} x_n\right)\right) \\
&< \lambda_1 \cdots \lambda_n
\end{aligned}$$

if for all i, we have $\lambda_i^{-1} x_i \in U_i$. For each $i \in [n]$, we have U_i is an open neighborhood of $0 \in V_i$, so there exists some continuous seminorm σ_i and $\varepsilon_i > 0$ such that $U(0, \sigma_i, \varepsilon_i) \subseteq U_i$. If for every i, we have $\sigma_i\left(\lambda_i^{-1} x_i\right) < \varepsilon_i$, then

$$\tau(M(x_1, \dots, x_n)) < \lambda_1 \cdots \lambda_n.$$

Rewriting, we have $\lambda_i > \frac{1}{\varepsilon_i} \sigma_i(x_i)$ for each such i, so that we have

$$\begin{aligned}
\tau(M(x_1, \dots, x_n)) &\leq \lambda_1 \cdots \lambda_n \\
&\leq \inf\left(\left\{\lambda_1 \cdots \lambda_n : \lambda_i > 0 \text{ and for all } i \in [n],\, \lambda_i > \frac{1}{\varepsilon_i} \sigma_i(x_i)\right\}\right) \\
&= \left(\frac{1}{\varepsilon_1} \sigma_1(x_1)\right) \cdots \left(\frac{1}{\varepsilon_n} \sigma_n(x_n)\right)
\end{aligned}$$

Setting $\rho_i = \frac{1}{\varepsilon_i} \sigma_i$ yields the desired result.

Assume that (†) holds. Let $(x_1, \dots, x_n), (y_1, \dots, y_n) \in V_1 \times \cdots \times V_n$. We want to try to compare the difference between $M(x_1, \dots, x_n)$ and $M(y_1, \dots, y_n)$. From multilinearity, we have

$$\begin{aligned}
M(x_1, \dots, x_n) - M(y_1, \dots, y_n) &= \sum_{i=1}^{n} (M(y_1, \dots, y_{i-1}, y_i, x_{i+1}, \dots, x_n) - M(y_1, \dots, y_{i-1}, x_i, x_{i+1}, \dots, x_n)) \\
&= \sum_{i=1}^{n} M(y_1, \dots, y_{i-1}, y_i - x_i, x_{i+1}, \dots, x_n)
\end{aligned}$$

Let $Z \subseteq W$ be open containing $z = M(x_1, \dots, x_n)$, and write $z' = M(y_1, \dots, y_n)$. Then, there exists a continuous seminorm τ on W and $\varepsilon > 0$ such that $U(z, \tau, \varepsilon) \subseteq Z$.

Consider $\tau(z - z')$. Then, we have by multilinearity and the triangle inequality that

$$\tau(z - z') \leq \sum_{i=1}^{n} \tau(M(y_1, \dots, y_{i-1}, y_i - x_i, x_{i+1}, \dots, x_n)).$$

Using (†) gives continuous seminorms $\rho_1, \dots, \rho_n$ on $V_1, \dots, V_n$ respectively such that

$$\leq \sum_{i=1}^{n} \rho_1(y_1) \cdots \rho_{i-1}(y_{i-1}) \rho_i(y_i - x_i) \rho_{i+1}(x_{i+1}) \cdots \rho_n(x_n)$$

$$\leq n(C+\eta)^{n-1}\eta,$$

where we assume that for all i, $y_i \in U(x_i, \rho_i, \eta)$, and $C = \max\{\rho_i(x_i) : i = 1, \dots, n\}$. By choosing η sufficiently small, $n(C+\eta)^{n-1}\eta < \varepsilon$, which yields

$$\begin{aligned} M\left(\prod_{i=1}^{n} U(x_i, \rho_i, \eta)\right) &\subseteq U(z, \tau, \varepsilon) \\ &\subseteq Z, \end{aligned}$$

so M is continuous. □

Notation

- The natural numbers are given by $\mathbb{N} = \{0, 1, 2, 3, \dots\}$.
- The strictly positive natural numbers are given by $\mathbb{Z}_{>0} = \{1, 2, 3, \dots\}$.
- A multi-index is a tuple $\alpha \in \mathbb{N}^n$. We let

  $$|\alpha| = \alpha_1 + \cdots + \alpha_n.$$

  If $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, then we will write

  $$x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$$

  and

  $$\alpha! = \alpha_1! \cdots \alpha_n!$$

  for the corresponding values.

  If $f \colon \mathbb{R}^n \to \mathbb{R}$ is any function, then we will write

  $$\partial^\alpha f = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}.$$

- If A is a set, we will write $\mathcal{P}(A)$ for the power set of A. If A and B are sets, then the set $B^A \subseteq \mathcal{P}(A \times B)$ is the set of all functions $f \colon A \to B$.

  We will write $\mathcal{P}_{\text{fin}}(X)$ to consist of all finite subsets subsets of X.
- If $x \in \mathbb{R}^n$ is given by $(x_1, \dots, x_n)$, we will define the bracket

  $$\langle x \rangle := \sqrt{1 + x_1^2 + \cdots + x_n^2}.$$

  In the particular case of $n = 1$, we will write $\langle k \rangle = \sqrt{1+k^2}$ for any $k \in \mathbb{N}$.
- If $n \in \mathbb{N}$, then

  $$[n] = \{1, 2, \dots, n\},$$

  where we have $[0] = \emptyset$.
- If A is a finite set, then $|A|$ is the cardinality of A.
- Generally we will denote by $\mathbb{K}$ for either $\mathbb{R}$ or $\mathbb{C}$.